Zenith Quest

Your Personal Sky Guide

Zenith Quest is an immersive celestial exploration platform designed as a dark-mode-first, premium stargazing companion. It aggregates real-time data from 7+ astronomical APIs and synthesizes them into personalized, interactive sky guides — using custom orbital calculations, Three.js celestial spheres, real-time maps, and gamified challenges.

Installation and Setup Instructions

Follow these steps to configure, install, and run Zenith Quest on your local development machine.

Prerequisites

- **Node.js** — v18.x or later recommended
- **npm** — or your preferred package manager (yarn, pnpm)
- **MongoDB** — a local running instance or a MongoDB Atlas URI

Step 1: Clone the Repository

Clone the codebase and navigate to the project root:

```
git clone <repository-url>

cd "Zenith Quest"
```

Step 2: Install Dependencies

Install all required Node.js libraries:

```
npm install
```

Step 3: Configure Environment Variables

Copy the environment template file to create your local configuration:

```
cp .env.example .env.local
```

Open .env.local in your text editor and fill in the following keys:

```
# Clerk Authentication (https://clerk.com)

NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=your_key

CLERK_SECRET_KEY=your_key


# MongoDB Connection String

MONGODB_URI=mongodb://localhost:27017/zenith-quest


# NASA API Key (free tier, defaults to DEMO_KEY)

NASA_API_KEY=your_key


# AstronomyAPI (https://astronomyapi.com)

ASTRONOMY_API_ID=your_id

ASTRONOMY_API_SECRET=your_secret


# N2YO Satellite Tracking (https://www.n2yo.com)

N2YO_API_KEY=your_key


# OpenAI API Key (optional — for AI Sky Narrator)

OPENAI_API_KEY=your_key


# Public Routing

NEXT_PUBLIC_APP_URL=http://localhost:3000
```

Note: Fully Functional Demo Mode — If you do not have active keys for N2YO, AstronomyAPI, or OpenAI, the application uses built-in mathematical astronomy calculators, keyless ISS tracking endpoints, and local deterministic narrative templates as comprehensive fallbacks.

Step 4: Run MongoDB

If you are using a local MongoDB database, start the daemon:

```
mongod
```

Step 5: Start the Development Server

Launch the local environment using Next.js Turbopack:

```
npm run dev
```

Open your browser and navigate to http://localhost:3000.

Website Functionality and Unique Features

Zenith Quest stands out by combining rich astronomical visualizations with context-aware gamification and real-time computation.

1. Proprietary Sky Score Engine

Calculates a composite 0–100 Sky Quality Score dynamically optimized for your local coordinates. Evaluates 6 weighted factors:

- **Cloud Cover (30%)** — Powered by Open-Meteo; lower is better.
- **Light Pollution (25%)** — Bortle Scale (1–9); lower is better.
- **Visible Planets (15%)** — Real-time planetary visibility; more is better.
- **Satellite Activity (10%)** — Count of satellite passes above you.
- **Moon Brightness (10%)** — Illumination level; critical for deep sky objects.
- **Transparency (10%)** — Atmospheric visibility in kilometers.

Outputs clear viewing recommendations (Excellent / Good / Average / Poor) and computes optimal hourly stargazing windows.

2. AI Sky Narrator

- **Natural-language narrative** synthesized from real-time weather and celestial position data.
- **Powered by OpenAI GPT-4o-mini** for concise daily sky summaries and viewing tips.
- **Smart deterministic fallback** constructs logical narratives locally when API keys are unavailable.

3. Interactive 3D Sky Dome

- **Three.js + React Three Fiber** — immersive celestial dome rendered in real-time.
- **Displays constellations, stars, planets, and the Moon** mapped to your exact latitude, longitude, and current local time.
- **Full interaction** — rotation, zoom, and click to inspect astronomical coordinates (RA, Dec, Altitude, Azimuth).

4. Real-Time Satellite Tracker

- **Live satellite orbits** visualized on a beautiful Leaflet map.
- **Tracks ISS, Hubble, weather satellites, and Starlink** constellations.
- **Calculates visual pass schedules**, maximum elevations, azimuth trajectories, and compass directions.

5. Gamified Missions & XP

- **3 mission types** — Observation, Tracking, and Identification challenges.
- **Progression system** — XP rewards, 7 explorer ranks (Stargazer !' Universe Sage), persistent observation history.
- **Badge reward system** — 5 rarity tiers across diverse observation achievements.

6. NASA Integration

- **Astronomy Picture of the Day (APOD)** — displayed with full captions.
- **Near-Earth Objects (NEOs)** — relative velocity, miss distance, and hazardous flags.
- **Solar Activity Monitoring** — flares, geomagnetic storms, solar wind speed, and Aurora forecasts via NOAA/NASA DONKI.

7. Celestial Timeline

Provides a detailed sunset-to-sunrise timeline showing planetary rise/set/transit times, twilight boundaries, and astronomical milestones.

Dependencies

Zenith Quest is built on a modern, highly optimized JavaScript/TypeScript ecosystem.

Core Framework & Build Tools

Dependency	Version	Purpose
Next.js	16	App Router, Turbopack, SSR, Route Handlers
React	19	Server Components, Suspense, Hooks
TypeScript	5	Strictly-typed schemas & service interfaces
Tailwind CSS	v4	Utility-first styling with CSS variables
Framer Motion	12	Page transitions and micro-animations

Astronomical & Visual Tools

Dependency	Purpose
Three.js	3D rendering engine for celestial visualizations
React Three Fiber	React renderer for Three.js scenes
Drei (@react-three/drei)	Helper components for React Three Fiber
Leaflet & React-Leaflet	Interactive maps for satellite coordinates
Recharts	Interactive dashboards and observation analytics

Backend & Data Infrastructure

Dependency	Purpose
MongoDB & Mongoose	NoSQL storage — profiles, badges, observation logs
Clerk (@clerk/nextjs)	Secure OAuth sign-in (Google & GitHub)
OpenAI Node SDK	LLM-based narrative generation
SWR	Data caching and revalidation for API fetchers
Zustand	Client-side global state management
PDFKit	Backend PDF generation for proposals

External APIs

API	Data Provided
Open-Meteo	Cloud cover, visibility, humidity, wind, forecasts
AstronomyAPI	Planetary coordinates, moon phase, illumination
N2YO	Satellite passes, trajectories, overhead arrays
NASA Web Services	APOD, NeoWs, Solar Flare DONKI
wheretheiss.at	Keyless real-time ISS tracking

Development Commands

Command	Purpose
npm run dev	Start dev server with hot module replacement
npm run build	Compile production-ready bundle
npm start	Run the built production application
npm run lint	Run ESLint checks